NeuroShot

Dynamic Trajectory Optimization for Moving Targets using DRL

Deep Reinforcement Learning: Project Proposal

Team Name: **Insight Engineers**

Project by

Niladri Ghosh	**Raihan Uddin**	**Sayan Das**
B2430100	B2430070	B2430035

Problem Description and Motivation

The Problem Statement

In traditional robotics and gaming environments, hitting a static target is a solved problem using basic trigonometry. However, when the target is in motion, the complexity increases exponentially. The system must not only calculate the required force and angle for a projectile but must also **predict the future position** of the target based on its current velocity and direction.

Specifically, this project addresses the challenge of a stationary agent attempting to intercept a moving goal. The ball is located at a fixed origin point $P(0, 0)$, while a basket moves within a predefined range $[x_{min}, x_{max}]$ at a variable velocity v_t. The agent must decide on the instantaneous launch velocity V_0 and launch angle θ to ensure the projectile path intersects with the basket's future coordinates.

Motivation

The motivation behind this project is to explore the intersection of **Physics-Engine Simulations** and **Deep Reinforcement Learning (DRL)**. Success in this simulation mimics real-world challenges in:

- **Automated Logistics:** Precise placement of items into moving bins on high-speed conveyor belts.
- **Defense & Aerospace:** Developing interception algorithms for moving objects in 3D space where traditional PID controllers might be too rigid.
- **Sports Analytics:** Creating intelligent training bots that can accurately simulate pass/shot dynamics in real-time.

Proposed Approach and Methodology

The system will be modeled as a Markov Decision Process (MDP) defined by the tuple (S, A, P, R).

State and Action Space

- **State Space (S):** The agent observes the current position of the basket (x_t, y_t), its velocity v_t, and its distance from the ball.
- **Action Space (A):** A continuous space consisting of two primary parameters:
  1. **Launch Force/Magnitude:** The initial speed V_0.
  2. **Launch Angle:** The angle θ relative to the horizontal plane.

Proposed AI Techniques

Since the action space is continuous, we will move away from discrete Q-Learning and utilize **Policy Gradient methods**. We plan to implement and compare two primary RL algorithms:

1. **Deep Deterministic Policy Gradient (DDPG):** An actor-critic, model-free algorithm for high-dimensional, continuous action spaces. The "Actor" proposes an angle and force, while the "Critic" evaluates the expected return.
2. **Proximal Policy Optimization (PPO):** Serving as our baseline, PPO is known for stability and prevents "catastrophic forgetting" by constraining policy updates.

Physics Simulation & Reward Design

The environment will utilize a physics engine (e.g., Pymunk or PyBullet). The reward function R is defined as:

$$R = \begin{cases} +100 & \text{if Ball enters Basket} \\ -d & \text{if Ball misses (where } d \text{ is the final distance)} \\ -1 & \text{per time step (to encourage speed)} \end{cases}$$

Expected Outcome and Deliverables

Deliverables

- **Trained RL Model:** A neural network capable of predicting required launch vectors across varying target speeds.
- **Interactive Simulation:** A visual environment (Pygame/Matplotlib) demonstrating the agent's learning progress.
- **Performance Analysis:** A report featuring learning curves and accuracy heatmaps across different trajectories.

Expected Application

The final application will demonstrate an autonomous system that adapts to different trajectories without manual physics re-coding, relying on its learned "intuition" of environment dynamics.

Tentative Work Division

Member	Primary Responsibilities
Niladri Ghosh	**Environment Development:** Designing physics simulation, target movement patterns, and state-space logic.
Raihan Uddin	**Algorithm Implementation:** Designing Actor-Critic architecture, hyperparameter tuning, and reward shaping.
Sayan Das	**Evaluation & Visualization:** Developing the visual interface, logging training data, and comparative analysis.